

Legible Consensus: Topology-Aware Quorum Geometry for Asymmetric Networks

Paper #98

Abstract

Quorum design over asymmetric topologies conflates two independent concerns: inter-tier obligation (which tiers must participate for cross-tier safety) and intra-tier replication (how each tier survives local failures). Flat quorums treat all nodes as interchangeable; when consensus fails, the structure does not reveal whether a tier was unreachable or a tier lost too many replicas. We show that mapping a crumbling-wall quorum construction to a physically tiered network separates these concerns and makes the protocol’s failure modes *legible*: an operator can determine which tiers retain global consensus capability from the wall structure and connectivity state alone, without trial quorum formation. Legibility extends from liveness to semantics: the wall and connectivity state jointly reveal which capabilities each tier retains—whether it can still *extend* the globally ordered log (complete both consensus phases) and whether it can still *learn* it (assemble acceptance evidence from a complete Phase 2 quorum)—and therefore which consistency guarantees the service can still offer at that tier. A tier retaining both capabilities continues operating through the global log; a cut-off tier is structurally confined to its local replica, and the wall says which, and when. Using a 10-node Earth/LEO/Moon/Mars topology as a magnifying glass, we confirm that three of four tiers retain global liveness during Mars conjunction blackout; only the disconnected tier loses it. Consensus latency at each tier tracks the speed-of-light critical path to Earth: 183 ms (Earth), 131 ms (LEO), 5.1 s (Moon). The wall also imposes a leadership cost gradient on Multi-Paxos elections that symmetric grid quorums cannot express. A comparison between sparse and full-coverage topologies separates wall obligations from network reachability as independent liveness constraints. All results are design-level; quorum intersection is verified exhaustively in TLA+.

1 Introduction

Consensus protocols are usually studied over networks where all participants are roughly equidistant. In practice, many systems span topologies with structured asymmetry: datacenters at different distances, edge nodes behind slow links, or, in the extreme case that motivates this paper, tiers

separated by minutes to hours of light-travel time.

The naive approach to consensus over such a topology is a flat quorum: majority (or any uniform threshold) over all nodes. A uniform threshold couples the commit path to node counts rather than structure: whenever the fast tier alone falls below the threshold, every operation waits on at least one slower tier—in our topology, a 6-of-10 majority leaves Earth on every attempt (Section 6 measures the cost)—and constructions that require every tier wait on the slowest one. Flexible Paxos [9] showed that safety requires only cross-intersection between Phase 1 and Phase 2 quorum families, not universal majority. This decouples the two phases and opens a design space: Phase 2 (the hot commit path) can be small and fast while Phase 1 (the rare election path) can be large and slow, as long as they intersect.

But how should that design space be explored? Most Flexible Paxos deployments choose quorum sizes, a number for each phase, without considering the *shape* of those quorums. Peleg and Wool [16] showed that quorum geometry can be engineered around combinatorial structure rather than simple counts, with crumbling-wall constructions (nodes arranged in rows of varying width; a quorum takes one complete row together with a representative from every row below it, and the wall “crumbles” as rows lose members to failure) that achieve high availability through asymmetric quorum families. This paper composes these two ideas: Flexible Paxos’s intersection requirement with Peleg and Wool’s wall construction, mapped onto a physically tiered network where the rows of the wall correspond to latency tiers. The composition itself was suggested by Howard et al.; our contribution is what the composition is shown to yield—per-tier liveness characterization under scheduled disconnection, legibility extended from liveness to per-tier consistency semantics, and a leadership cost gradient.

The deeper problem with flat quorums is not performance but *conflation*. Quorum design over asymmetric topologies involves two independent concerns: *inter-tier obligation* (which tiers must participate in each phase for cross-tier safety) and *intra-tier replication* (how each tier internally survives node failures for local durability). A flat quorum treats every node as an interchangeable member of a single obligation pool, regardless of the structural role its tier plays. Nodes separated by minutes of light-travel time are placed in

the same quorum as nodes across the room. When a quorum fails to form, the failure is opaque—was it a coverage problem (a tier is unreachable) or a replication problem (too many nodes crashed within a tier)? The structure doesn’t say.

The wall separates these concerns. Inter-tier obligation is encoded in the wall geometry: one witness per tier on the downward path, with safety anchored at the bottom row. Intra-tier replication is each non-anchor tier’s own concern, invisible to the wall; the anchor row is the exception, since its replication threshold is itself the Phase 2 quorum (Section 7). This separation produces a protocol whose failure modes are *legible*. We call a quorum construction legible if, once the current connectivity state is known, an operator can determine which tiers retain global consensus capability by inspecting the wall structure against a tier-level connectivity summary—an $O(\text{tiers})$ check rather than trial quorum formation or enumeration over quorum subsets. (Acquiring the connectivity state itself requires failure detection or, in our setting, orbital schedule knowledge; legibility concerns what follows once it is known.)

We demonstrate this using Eidolon, a discrete-event Paxos simulator, over a 10-node topology spanning Earth (5 nodes), LEO (1), Moon (1), and Mars (3)—the 5/1/1/3 topology. The interplanetary setting is not the point; it is the magnifying glass. At 22 light-minutes, the conflation error in flat quorum design is physically undeniable; the same error exists at terrestrial scales but is masked by generous timeouts.

This paper makes five contributions:

1. **The crumbling-wall construction mapped to physical topology.** Per-tier Phase 1 quorum families where each tier reads downward through the wall. Phase 2 is the Earth row. Intersection holds because every Phase 1 contains at least one Earth node. The construction is verified exhaustively in TLA+ over the full 10-node topology (Section 3).
2. **Legible degradation of liveness and consistency.** Under Mars conjunction blackout, the wall crumbles from the top: only Mars loses global Phase 1 capability. Earth, LEO, and Moon retain it. Consensus latency at each tier tracks the speed-of-light critical path to Earth: 183 ms (Earth), 131 ms (LEO), 5.1 s (Moon). The same structure determines semantics: a tier that can complete both phases under current connectivity continues operating through the globally ordered log; a tier that can still assemble acceptance evidence from a complete Phase 2 quorum can learn the global history even when it cannot extend it; a tier that can do neither is confined to its local replica (Section 9). Section 6 includes a flat-vs-wall calibration on the same topology.
3. **Wall obligations versus network reachability.** A sparse network topology where LEO sees only 3 of 5 Earth ground stations breaks LEO’s liveness despite the

wall permitting it. The wall determines quorum *obligations*; the network determines *achievability*. These are independent constraints that compose to determine liveness (Section 6.3).

4. **Crash tolerance and the weakest-link migration.** Relaxing Phase 2 from all-of-Earth to k -of-Earth introduces crash tolerance, but the weakest link migrates from the global quorum to the Earth-local quorum. Under two Earth crashes, the topology-aware global construction maintains 100% during-blackout success while the Earth-local construction loses liveness outright (Section 7).
5. **Leadership cost gradient.** The wall imposes a monotonic cost hierarchy on Multi-Paxos leader election: a closed-form count, confirmed by exhaustive enumeration, gives Earth $4.6\times$ more valid Phase 1 quorums than Mars, and the per-tier families are nested, so every crash pattern survivable from a higher tier is survivable from every tier below it. Symmetric grid quorums cannot express this gradient (Section 9).

Each empirical claim is tied to an executable artifact and output file (Appendix A). All results are design-level; deployment would introduce orbital dynamics, antenna scheduling, and variable link quality not modeled here.

2 Background

2.1 Flexible Paxos

Classic Paxos [11] uses majority quorums for both phases, which are Phase 1 (prepare/promise) and Phase 2 (accept/accepted), guaranteeing intersection via the pigeonhole principle.

Flexible Paxos [9] observed that the intersection requirement is *cross-phase*, not within-phase: any Phase 1 quorum must intersect any Phase 2 quorum, but Phase 1 quorums need not intersect each other, nor Phase 2 quorums each other. Formally, safety holds if $q_1 + q_2 > |N|$, where q_1 and q_2 are the quorum sizes for Phase 1 and Phase 2 respectively. This is the uniform special case where all quorums have the same size; the crumbling-wall construction generalizes this with per-tier quorum families of different sizes, where the cross-intersection property holds by construction rather than by arithmetic majority. The key consequence is that Phase 2 (the hot path, executed on every commit) can be small if Phase 1 (the cold path, executed only during leader election) is correspondingly large.

2.2 Crumbling-Wall Quorum Systems

Peleg and Wool [16] introduced crumbling walls as quorum constructions that achieve high availability through asym-

metric structure. Nodes are arranged in rows of potentially different widths. A quorum takes one complete row together with one representative from every row below it. The “crumbling” refers to rows losing members to failure.

The key property: any two quorums intersect. If their complete rows differ, the quorum whose complete row sits higher carries a representative in the other quorum’s complete row; if they share a complete row, they share it outright—the intersection that consensus requires.

2.3 Composing the Two Ideas

Grid and crumbling-wall constructions [3, 16] show that quorum geometry can be structured around combinatorial properties rather than simple counts. Flexible Paxos shows that the two phases of consensus can use different quorum families. Our construction composes these: the rows of the crumbling wall correspond to physical tiers of the network, and each tier’s Phase 1 quorum family reads downward through the wall from that tier’s position. Phase 2 uses only the bottom row (the fastest tier). The result decomposes wall-like structure across the two phases of Flexible Paxos rather than reproducing Peleg and Wool’s construction, whose quorums take one complete row plus a representative of every row below it: here the downward witness chain encodes participation policy in Phase 1, while the complete anchor row provides Phase 2 safety. Only the anchor intersection is safety-critical; the upper-tier witnesses exist to make authority and degradation topology-legible, not to secure agreement.

3 Construction

3.1 System Model

Let the acceptor universe be

$$N = E \cup L \cup U \cup M,$$

where the tiers are pairwise disjoint and represent Earth, LEO, Moon, and Mars respectively. In the evaluated topology,

$$|E| = 5, \quad |L| = 1, \quad |U| = 1, \quad |M| = 3, \quad |N| = 10.$$

We call this the 5/1/1/3 topology. Tiers are ordered by latency: Earth is the fastest (bottom of the wall), Mars is the slowest (top). The network is asynchronous with configurable per-link delay and jitter. Acceptors are crash-stop and non-Byzantine. Links may be removed during scheduled blackout windows.

The simulator uses three logical consensus scopes:

- **Earth-local:** Flexible Paxos over the five Earth nodes ($q_1 = 4, q_2 = 2$).

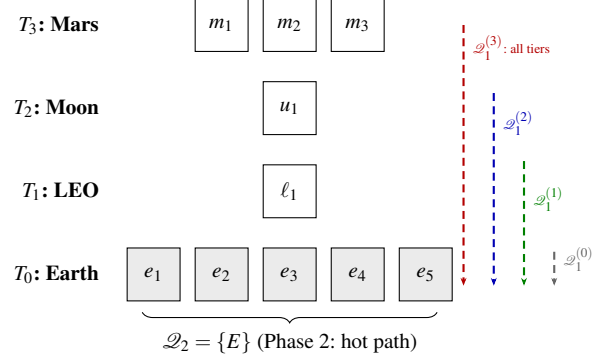


Figure 1: The crumbling-wall construction over the 5/1/1/3 topology. Each tier’s Phase 1 quorum reads downward through the wall (dashed arrows). Phase 2 uses only the Earth row (shaded). The wall narrows from bottom to top: wider rows offer more quorum choices.

- **Mars-local:** majority quorum over the three Mars nodes ($q = 2$).
- **Global:** the crumbling-wall construction defined below, over all ten nodes.

3.2 Per-Tier Phase 1 Families

The wall maps tiers to rows, ordered bottom (Earth, fast) to top (Mars, slow), as shown in Figure 1. We index tiers from the bottom: $T_0 = E$ (Earth), $T_1 = L$ (LEO), $T_2 = U$ (Moon), $T_3 = M$ (Mars). A proposer at tier i forms its Phase 1 quorum by reading *downward* from row i : it needs at least one respondent from its own tier and every tier below.

Initiating tier	Phase 1 scope	Tiers	Min size
Earth (tier 0)	Earth only	1	1
LEO (tier 1)	LEO + Earth	2	2
Moon (tier 2)	Moon + LEO + Earth	3	3
Mars (tier 3)	all four tiers	4	4

Let $\mathcal{Q}_1^{(i)}$ denote the Phase 1 family for a proposer at tier i :

$$\mathcal{Q}_1^{(i)} = \{Q \subseteq N \mid \forall j \leq i: Q \cap T_j \neq \emptyset\}.$$

A proposer at tier i needs at least one node from every tier at or below it ($j \leq i$). For example, $\mathcal{Q}_1^{(2)}$ (Moon) requires intersection with T_0 (Earth), T_1 (LEO), and T_2 (Moon), which is every tier from Moon down to Earth. For relaxed Phase 2 (Section 7), Phase 1 additionally requires enough Earth nodes to satisfy a pigeonhole intersection constraint.

3.3 Phase 2 Family

The Phase 2 family places the hot commit path entirely on the fast tier:

$$\mathcal{Q}_2 = \{E\}.$$

That is, Phase 2 requires all five Earth nodes (the strict construction). Section 7 relaxes this to k -of- $|E|$.

3.4 Safety: Quorum Intersection

Proposition 1. For every tier i and every $Q_1 \in \mathcal{Q}_1^{(i)}$, $Q_2 \in \mathcal{Q}_2$: $Q_1 \cap Q_2 \neq \emptyset$.

Proof. Every $\mathcal{Q}_1^{(i)}$ requires $Q_1 \cap T_0 \neq \emptyset$ (since $0 \leq i$ for all i , i.e., Earth is at or below every tier). $Q_2 = E = T_0$. Therefore $Q_1 \cap Q_2 \supseteq Q_1 \cap T_0 \neq \emptyset$. $\square$

The intersection follows from the wall’s structure: every path through the wall reaches the bottom row, and Phase 2 is the bottom row.

We verified this mechanically in TLA+. For readers who have not used it: TLA+ is a specification language whose model checker, TLC, enumerates *every* state reachable from the initial conditions of a finite model and checks the stated invariant in each one. A run is *complete* when that enumeration reaches a fixed point—no reachable state was left unexamined—so a complete run is a proof of the invariant for that finite model, not a statistical sample of behaviors. The QuorumIntersection specification enumerates every (construction $\times$ initiating tier $\times Q_1 \times Q_2$) combination for the strict, $k=4$, and $k=3$ families over the full 10-node topology (27,921 states, complete); ExhaustiveIntersection re-checks the strict and $k=4$ families as parse-time assumptions. No intersection failure was found.

For the full Paxos protocol (not just quorum intersection), we verify safety over a reduced 6-node topology ($3E + 1L + 1U + 1M$) that preserves the quorum-intersection structure. The invariant the reduction preserves is the pigeon-hole inequality, not tier membership alone: in the reduced model both quorum families guarantee at least two of the three Earth nodes ($2 + 2 > 3$), and in the full modeled family Q_1 carries at least two Earth nodes against a 4-of-5 Q_2 ($2 + 4 > 5$)—both instantiate $|Q_1 \cap E| \geq |E| - k + 1$ with the margin $|E| - k = 1$ held fixed. (Tier membership alone suffices only under strict $\mathcal{Q}_2 = \{E\}$, where any Earth node in Q_1 intersects it.) Liveness properties (crash tolerance, progress under partial failure) are topology-size-dependent and are evaluated only at full scale. Over the reduced topology, single-decree Paxos agreement was model-checked completely (67M states, no violation).

Verification scope. The protocol-level specification models a single all-tier Phase 1 family rather than the per-tier families of Section 3: it exercises the protocol machinery under wall-shaped quorums, but safety of the per-tier construction itself rests on Proposition 1 and the standard Flexible Paxos argument, with the intersection premise checked exhaustively above. The proof is primary; the model checking is corroboration. Specifications are in tla/ (see Appendix A).

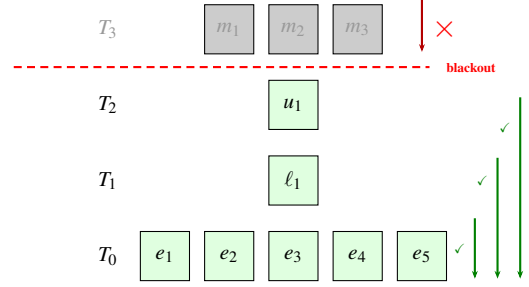


Figure 2: Liveness under Mars conjunction blackout. The partition (dashed line) disconnects T_3 . Earth, LEO, and Moon can still read downward to form Phase 1 quorums (green arrows). Mars cannot reach any inner tier (red $\times$). The wall crumbles from the top.

3.5 Liveness: Reading the Wall

The wall determines liveness as follows. A proposer at tier i can complete Phase 1 if and only if it can reach at least one node in every tier j where $j \leq i$ (from its own tier down to Earth). Phase 2 succeeds if the proposer can reach enough Earth nodes (all five under strict $\mathcal{Q}_2$).

During Mars conjunction blackout, cross-tier links between Mars and the inner system are removed. The connectivity state is:

- Mars $\leftrightarrow$ {Moon, LEO, Earth}: **partitioned**.
- Moon $\leftrightarrow$ LEO $\leftrightarrow$ Earth: **connected**.

Reading the wall against this connectivity:

- **Earth** ($i=0$, needs T_0): **works**. Trivially satisfied.
- **LEO** ($i=1$, needs T_0, T_1): **works**. Both tiers reachable.
- **Moon** ($i=2$, needs T_0, T_1, T_2): **works**. All three tiers reachable.
- **Mars** ($i=3$, needs T_0, T_1, T_2, T_3): **blocked**. Cannot reach inner tiers.

The liveness failure is scoped to exactly the disconnected tier. The wall crumbles from the top (Figure 2).

This instantiates the legibility definition of Section 1: for any tier i , global Phase 1 succeeds if and only if every tier $j \leq i$ is reachable from i in C —the promised $O(\text{tiers})$ check. This is what distinguishes legibility from mere computability. The claim concerns quorum formability: it bounds what a sole, timely proposer can achieve under connectivity state C , and does not speak to contention between concurrent proposers, whose durable promises can delay an otherwise-capable peer. Global *operation*, as opposed to Phase 1 recovery, additionally requires the Phase 2 quorum to be reachable; Section 6.3 shows the two conditions coming apart.

3.6 What “Global” Means

When an Earth-based proposer completes both phases using only Earth nodes, the decision is *global* in the Paxos sense: any future proposer at any tier, upon completing Phase 1, will discover this value in the Earth acceptors’ state and cannot contradict it. Global consensus does not require that all tiers participated, only that no future quorum can produce a conflicting result.

4 Related Work

Paxos established the standard safety argument for replicated state machines under asynchronous messaging and crash failures [11]. Flexible Paxos showed that safety requires only Phase 1/Phase 2 cross-intersection, not universal majority [9]. Our construction builds directly on that result; the novelty is not in the intersection theorem but in mapping the quorum shape to physical topology and characterizing the resulting liveness properties per tier.

Grid and crumbling-wall quorum constructions [3, 16] demonstrate that quorum geometry can be engineered around combinatorial structure. Howard et al. note explicitly that Flexible Paxos can use such constructions; we take up that suggestion. Peleg and Wool analyzed crumbling-wall availability under random, independent node failures. Our contribution is not the wall itself but the mapping: rows correspond to physical latency tiers, and the resulting per-tier liveness properties under scheduled, topology-correlated disconnection, are the properties the abstract framework identifies as possible but does not explore. Li, Chan, and Lesani [12] formalize heterogeneous quorum systems in which each process may use a different quorum family, and give necessary and sufficient conditions for liveness; our per-tier Phase 1 families are a concrete, topology-aligned instantiation of that general idea. Their Satrapy system [13] carries the framework from abstract quorum systems to practical consensus. Hierarchical quorum structures have a longer history in database replication [5] and coordination services; our wall differs in that the hierarchy reflects physical latency tiers rather than organizational or logical grouping.

Geo-distributed consensus provides the closest practical analog. WPaxos [1] uses Flexible Paxos with grid quorum layouts and per-object multi-leader coordination to keep commits local; it is the nearest point of comparison. WPaxos’s grid quorums are symmetric: Q_1 spans $Z - f_z$ zones and Q_2 spans $f_z + 1$ zones, with the same structure regardless of which node initiates. Our per-tier $\mathcal{Q}_1^{(i)}$ families are asymmetric as the initiating tier determines the quorum scope, which produces a leadership cost gradient that symmetric grids structurally cannot express (Section 9). EPaxos [15] removes the leader bottleneck for non-conflicting commands. Mencius [14] distributes leader re-

sponsibility across WAN sites. MDCC [10] achieves single-round cross-datacenter commits. All four optimize for tens-to-hundreds of milliseconds WAN latency and assume all replicas are usually reachable. Our setting differs in two respects: the asymmetry spans three orders of magnitude (sub-second to 1342 s), and the target phenomenon is scheduled total disconnection rather than heterogeneous latency.

Delay-tolerant networking [2, 7] motivates the operating regime but addresses routing and custody transfer, not consensus. CRDTs have been studied in DTN-like settings [8]. SwarmDAG [18] uses extended virtual synchrony for partition-tolerant robot swarms; Tran et al. [17] measure consensus latency under partitioning. Protocol Labs’ IPC [6] uses hierarchical subnets for blockchain scaling. NASA’s DSN study [4] evaluates database replication for ground stations. Our work occupies a gap: Paxos safety under topology-shaped quorum geometry and scheduled disconnection, with per-tier liveness characterization.

5 Experimental Design

Eidolon is a discrete-event simulator built on SimPy. Acceptors process Prepare and Accept messages with configurable processing delay. Proposers fan out messages to all acceptors and collect responses with a per-phase timeout. The network model assigns entities to named locations with per-link latency, jitter, and optional loss. Blackout is modeled as deterministic link removal; the repeater model substitutes a degraded alternate path. Event ordering is by simulation timestamp; the simulator is deterministic given a seed.

5.1 Topology

Five Earth ground stations (NA-West, Europe, Asia, SA-East, Africa) with cross-continental latencies of 30–120 ms. One LEO satellite with 20–45 ms links to ground stations. One Moon node at 1.28 s one-way. Three Mars nodes at configurable one-way delay (186–1342 s, representing closest approach to opposition). Mars-local links are 5 ms.

We evaluate two network variants:

- **Sparse:** LEO has links to 3 of 5 Earth stations (NA-West, Europe, Asia). Mars has links to 2 of 5 (NA-West, Europe). This models a single LEO satellite with limited ground coverage and a Mars relay with two tracking stations.
- **Full coverage:** LEO and Mars have links to all 5 Earth stations. This models a LEO constellation with global coverage and a deep-space network with worldwide antenna placement.

Table 1: Experimental design.

Experiment	Purpose	Key parameters
Parameter sweep	Confirm structural invariance of Earth-initiated global consensus across Mars delay and blackout duration	Mars delay $\in \{186, 750, 1342\}$ s; blackout $\in \{300, 900, 1800\}$ s; 50 seeds per point
Per-tier sweep	Measure which tiers retain global liveness; compare sparse vs. full-coverage topologies	Initiating tier $\in \{\text{Mars, Moon, LEO, Earth}\}$; both topologies; 186 s Mars delay; 900 s blackout; 50 seeds
Crash tolerance	Characterize the tradeoff between Phase 2 relaxation (k -of- $ E $) and crash tolerance	$k \in \{3, 4, 5\}$; 0–2 Earth crashes; 186 s; 900 s blackout; 50 seeds

5.2 Experiments

Three experiment families probe different aspects of the construction.

Fixed parameters across all experiments are listed in Table 2. All seeds are 40–89 (50 runs per point). Results are reported as means with 95% confidence intervals.

6 Results

6.1 Flat versus Crumbling Wall: Geometry Alone Changes the Outcome

To calibrate the effect of quorum geometry, we compare three constructions over the identical topology, blackout model, and parameter sweep. The *flat* construction uses a single Phase 1 family requiring all four tiers:

$$\mathcal{Q}_1^{\text{flat}} = \{Q \subseteq N \mid \forall j \in \{0, 1, 2, 3\} : Q \cap T_j \neq \emptyset\}.$$

Any blackout that removes a tier makes this quorum unformable. The *majority* baseline is a competitive shape-free Flexible Paxos configuration: $\mathcal{Q}_1^{\text{maj}} = \{Q \mid |Q| \geq 6\}$, any majority of the ten nodes, with no tier structure. The *crumbling wall* construction uses the per-tier Phase 1 families defined in Section 3, with an Earth-based global proposer. All three satisfy cross-intersection with $\mathcal{Q}_2 = \{E\}$: the tiered families require $Q_1 \cap T_0 \neq \emptyset$ directly, and any 6 of 10 nodes must include an Earth node because only five non-Earth nodes exist.

The flat construction yields 0% during-blackout success at every sweep point because it requires all tiers, including disconnected Mars. The crumbling wall yields 100% because the Earth-based proposer’s Phase 1 reads only Earth. Zero variance across 50 seeds confirms that outcomes here are structural consequences of quorum geometry and link state, not scheduler artifacts. We therefore treat this as calibration, not discovery.

Table 2: Simulator parameters.

Parameter	Value	Notes
Earth cross-cont. delay	30–120 ms	per link pair
LEO–ground delay	20–45 ms	per ground station
Moon one-way delay	1.28 s	fixed
Mars one-way delay	186–1342 s	swept
Mars-local delay	5 ms	fixed
Processing delay	1 ms	per acceptor
Jitter	$\pm 10\%$	of link delay, uniform
Per-phase timeout	500 s / 1 s	global / tier-local; budget raised to $1.25 \times$ the worst-case request-response path when that exceeds the base (above ≈ 200 s Mars delay)
Max Paxos rounds	1	per attempt
Blackout start	600 s base	pre-window scaled to fit one full global round plus margin (930 s at 186 s Mars delay)
Simulation end	4000 s base	horizon extends when pre/post windows scale
Reconciliation interval	120 s	between global attempts
Loss model	none	deterministic link removal for blackout

The majority baseline separates what the wall’s geometry buys from what any sufficiently slack quorum buys. Majority Phase 1 also sustains 100% during-blackout success for the Earth-initiated proposer: the blackout removes three nodes and a 6-of-10 quorum tolerates four, so blackout survival here is within the slack of a shape-free quorum. What the geometry changes is everything else. Two-phase decision latency doubles (0.361 s vs. 0.183 s, uniformly across the sweep): the sixth-fastest respondent lies outside Earth, so majority Phase 1 leaves the fast tier even when nothing has failed. Both constructions share the identical all-Earth Phase 2, so steady-state Multi-Paxos commits do not differ; the penalty falls on authority acquisition, recovery, and single-decree decisions—every operation that pays Phase 1. The tolerance is not tier-aware: during blackout a LEO-initiated proposer reaches only five acceptors (itself, three ground stations, and Moon), so majority Phase 1 is unformable from LEO while the wall’s LEO family still completes Phase 1. And the valid-quorum count collapses to a single tier-independent value ($\sum_{k \geq 6} \binom{10}{k} = 386$), below Earth’s 992 wall quorums and without the per-tier gradient of Figure 3. The wall’s contribution over a well-chosen flexible quorum is therefore not raw blackout survival, which slack

Table 3: Effect of quorum geometry: flat vs. majority vs. crumbling-wall Phase 1 over the same topology and blackout model. Earth-initiated global proposer, hard blackout, 50 seeds per point; rates are identical at all 9 parameter points (3 Mars delays $\times$ 3 blackout durations). Latency is the full two-phase (Phase 1 + Phase 2) decision latency of a single attempt.

Construction	During	Post	Latency
Flat Phase 1 (all tiers required)	0.0%	varies	varies
Majority Phase 1 (any 6 of 10)	100.0%	100.0%	0.361 s
Crumbling wall (Earth reads Earth only)	100.0%	100.0%	0.183 s

Table 4: Per-tier global consensus during Mars conjunction blackout (full-coverage topology, hard blackout, 186 s Mars delay, 900 s blackout, mean $\pm$ 95% CI over 50 seeds). Rates are identical across seeds (structural outcomes; no CIs shown); recovery lags are cadence artifacts (see text).
[†] Undefined: no Mars attempt lies wholly within this window (see text).

Tier	Position	During	Latency	Rec. lag
Earth	bottom (0)	100.0%	0.183 s	68.1 s
LEO	tier 1	100.0%	0.131 s	67.2 s
Moon	tier 2	100.0%	5.131 s	27.1 s
Mars	top (3)	— [†]	752.5 s	1009.3 s

alone can purchase; it is what slack cannot express—Phase 1 kept inside the fast tier, per-tier legibility of degradation, and the leadership gradient of Section 9.

6.2 Per-Tier Liveness: The Wall Crumbles from the Top

Table 4 reports the central result. Under full network coverage, each tier runs its own global proposer during Mars conjunction blackout (186 s one-way, 900 s blackout, hard blackout model, 50 seeds).

Three of four tiers retain global consensus capability during hard Mars blackout. Only Mars, which is the disconnected tier, loses it. The liveness failure crumbles from the top of the wall, exactly as the construction predicts. (95% CIs: latency within ± 0.001 s and recovery lag within ± 0.01 s for Earth, LEO, and Moon; ± 0.7 s and ± 1.7 s for Mars.)

The latency column maps directly to physics. Moon’s 5.131 s is two Paxos phases at the 1.28 s one-way Moon–Earth light delay ($\approx 4 \times 1.28$ s). Earth’s 183 ms reflects cross-continental round-trips within the Earth tier. LEO’s 131 ms is *faster* than Earth, which is a counterintuitive result explained by the network topology: LEO-to-ground links (20–45 ms) are shorter than cross-continental Earth links (50–120 ms). Wall position determines quorum *obligations*; the speed of light determines their *cost*. These are correlated but not identical, and LEO is the proof.

Mars fails during blackout for the structural reason pre-

Table 5: Per-tier global consensus under sparse topology (same parameters as Table 4). All rates are identical across the 50 seeds—structural outcomes, reported without CIs.

Initiating tier	Wall prediction	Sparse result
Earth	works	100.0%
LEO	works	0.0%
Moon	works	100.0%
Mars	blocked	0.0%

dicted by the wall (it cannot reach inner tiers for Phase 1). At the table’s 900 s point, however, no Mars attempt lies wholly within the blackout—a failed attempt occupies one 500 s Phase 1 timeout, and the attempt cadence, phase-shifted by the long pre-blackout round, straddles the boundary—so the during-blackout cell is undefined rather than zero. The prediction is observed directly at the 1800 s blackout duration, where the cadence places complete failed attempts inside the window: during-blackout success is 0% across all 50 seeds. Outside blackout, however, Mars-initiated global consensus *succeeds*: every pre- and post-blackout attempt completes across all 50 seeds, at a mean latency of 752.5 s—two Paxos phases at the 186 s one-way Mars–Earth delay ($\approx 4 \times 186$ s). Success requires a per-phase timeout budget sized to the initiating tier’s physics: the worst-case request–response path from Mars is $2 \times 186 \approx 372$ s, within the 500 s per-phase budget of Table 2. This is not a blackout finding but a physics finding: Mars-initiated global consensus is possible but pays the full interplanetary round trip on every phase, so in practice delegation to a lower-tier proposer dominates.

Recovery lag differences between tiers (27.1 s for Moon vs. 68.1 s for Earth) are cadence-alignment artifacts, not protocol properties. Recovery lag measures the gap from blackout end to the next successful cadence-aligned attempt, so it is quantized by the 120 s reconciliation interval: its value is bounded by one interval plus a round time, and where it falls within that bound depends only on where the blackout boundary lands in each tier’s attempt cadence.¹

6.3 Wall Obligations versus Network Reachability

Table 5 repeats the per-tier experiment under the sparse network topology.

LEO drops from 100% (full coverage) to 0% (sparse). The wall says LEO needs LEO + Earth; that obligation is satisfiable. But strict Phase 2 requires all five Earth nodes, and LEO has links to only three ground stations. The wall determines what a tier *needs*; the network determines what it *can*

¹ Each tier’s per-round time shifts its cadence phase relative to blackout end, which is why the values differ across tiers and move when the experiment window shifts. The effect is deterministic and independent of the consensus protocol; the bound, not the value, is the protocol property.

reach. Liveness requires both.

This makes legibility a two-step operator procedure: check the wall structure (which determines quorum obligations per tier) and check the network topology (which determines whether those obligations are achievable). The wall is the blueprint; the network is the building site.

Moon succeeds in both topologies because it has links to all five Earth ground stations. The sparse topology models a single LEO satellite with limited ground coverage; the three missing ground-station links are the difference between 0% and 100% for LEO-initiated consensus.

7 Crash Tolerance and Coordinated Relaxation

The strict Phase 2 choice $\mathcal{Q}_2 = \{E\}$ introduces a fragility: a single crashed Earth acceptor blocks all global Phase 2 progress. This section relaxes that choice.

7.1 Relaxed Construction

Let the relaxed Phase 2 family require k of $|E|$ Earth nodes. We evaluate $k = 4$ (tolerates one crash) and $k = 3$ (tolerates two). Phase 1 requires correspondingly more Earth nodes to maintain intersection: a k -of- $|E|$ Phase 2 quorum omits at most $|E| - k$ Earth nodes, so any set of $|E| - k + 1$ Earth nodes intersects every Phase 2 quorum (pigeonhole). Phase 1 must therefore contain at least $|E| - k + 1$ Earth nodes. For $k = 4$ out of 5 Earth nodes, this means at least 2 Earth nodes in every Q_1 ; for $k = 3$, at least 3. Both relaxed constructions are verified exhaustively in TLA+ alongside the strict one (Section 3).

The relaxed Phase 1 family for tier i is:

$$\mathcal{Q}_{1,k}^{(i)} = \{Q \subseteq N \mid \forall j \leq i: Q \cap T_j \neq \emptyset, |Q \cap E| \geq |E| - k + 1\}.$$

7.2 Results: The Weakest Link Migrates

Table 6 reports the crash-tolerance sweep (Earth-initiated, repeater-assisted, 186 s Mars delay, 900 s blackout, 50 seeds).

Strict Phase 2 is completely crash-intolerant. A single Earth crash produces permanent global liveness loss (row 2).

The weakest link migrates. At two crashes with standard Earth-local quorums, the *global* construction ($k = 3$) maintains 100% during-blackout success but the *Earth-local* construction loses liveness outright: it cannot form a Phase 1 quorum of 4 with only 3 remaining Earth nodes, so every attempt after the crashes land fails, and the 60.5% aggregate rate reflects only attempts completed before the crashes (rows 5–6). The global construction’s advantage is

not that it spans more nodes—for the Earth-initiated proposer used here, both phases of the relaxed global construction touch only Earth nodes—but that its Earth requirements are smaller: at $k = 3$, both global phases need only the three surviving Earth nodes, while the Earth-local configuration still demands four. The structural insight is the legibility of the migration: which crashes matter depends on which quorum family’s requirements they intersect, and the wall makes each family’s requirements explicit.

Coordinated relaxation restores resilience. Relaxing the Earth-local quorum to majority ($q_1 = q_2 = 3$) restores 100% local success with no loss of global success (row 6). The tradeoff curve, with Phase 1 minima shown for the bottom-initiated (Earth) and top-initiated (Mars) proposers:

Regime	P1 min (E/M)	P2 Tol.	Earth-local P2 quorum
Strict ($k = 5$)	1 / 4	5-of-5	0 2-node E
Relaxed ($k = 4$)	2 / 5	4-of-5	1 2-node E
Relaxed ($k = 3$) + maj	3 / 6	3-of-5	2 3-node E

An Earth-initiated Q_1 needs only the $|E| - k + 1$ Earth nodes; a Mars-initiated Q_1 adds one witness from each of the three upper tiers. Relaxation buys crash tolerance by growing the cold path, never the hot one: Phase 2 shrinks as Phase 1’s Earth requirement grows.

8 Threats to Validity and Limitations

All results are design-level, not deployment-level.

Anchor concentration. The wall concentrates both safety and liveness at the bottom row. Every Phase 1 reads down to Earth; every Phase 2 is on Earth. In our construction, this concentration is sufficient for the legibility property: per-tier liveness determination is $O(\text{tiers})$ because a fixed reference point for cross-intersection eliminates the need to enumerate quorum subsets. Distributing the anchor across multiple rows would require reasoning about which row combinations provide intersection—structured multi-anchor families might remain efficiently analyzable, but the wall’s simple linear readout would be lost. The single anchor is what makes the wall readable at a glance. Note, however, that the anchor is a *tier*, not a node. Since safety depends only on tier membership (Section 3), the anchor tier’s internal replication is unconstrained by the wall—it can be scaled, geo-distributed, or backed by its own consensus protocol independently. The threat is tier-level partition (all of Earth unreachable simultaneously), not individual node failure; relaxed Phase 2 tolerates the latter.

“Tier” is a bundle. A tier in this paper bundles several properties that happen to coincide in the planetary setting: a latency regime, a failure and partition domain, and, implicitly, an administrative domain. They need not coincide in general—two lunar bases operated by different agencies

Table 6: Coordinated relaxation under Earth crashes (mean $\pm$ 95% CI, 50 seeds). Earth-local Flexible Paxos: “std” is $q_1=4, q_2=2$; “maj” is $q_1=q_2=3$. Recovery is the lag from blackout end to the next successful global attempt and is quantized by the 120 s reconciliation cadence (Section 6).

Global $\mathcal{Q}_2$	Local	Crashes	Earth-local	Global during	Global post	Recovery (s)
strict	std	0	100.0%	100.0%	100.0%	93.11 $\pm$ 0.01
strict	std	1	100.0%	0.0%	0.0%	n/a
4-of-5	std	0	100.0%	100.0%	100.0%	94.27 $\pm$ 0.01
4-of-5	std	1	100.0%	100.0%	100.0%	94.26 $\pm$ 0.01
3-of-5	std	2	60.5%	100.0%	100.0%	94.64 $\pm$ 0.01
3-of-5	maj	2	100.0%	100.0%	100.0%	94.63 $\pm$ 0.01

would share a latency regime but not an administrative one, and two datacenters in one administrative domain may sit in different latency regimes. The wall as defined indexes only latency-ordered reachability; decomposing the remaining dimensions is future work.

Abstract network model. Blackout is deterministic link removal. Orbital dynamics, antenna scheduling, contention, signal coding, and variable link quality are not modeled. The LEO topology variant demonstrates that network reachability matters; a more realistic model with time-varying LEO coverage (orbital ground tracks) would show liveness fluctuating with satellite position.

Stylized workload. Reconciliation runs on a fixed 120 s interval. This creates scheduling artifacts in recovery lag measurements; recovery lag differences between tiers are workload properties, not protocol properties.

Single topology. We evaluate one topology size (5/1/1/3). The construction generalizes to other tier sizes, but we do not sweep Earth-tier size or Mars replication degree.

Crash-stop only. Byzantine behavior, correlated faults, reconfiguration during blackout, and storage failures are outside scope.

Per-tier crash tolerance. The interaction between wall position and crash tolerance (does a Moon-initiated proposer degrade differently from an Earth-initiated one under Earth crashes?) is not explored and remains future work.

9 Discussion

Separation of inter-tier obligation from intra-tier replication. The separation introduced in Section 1 has a concrete cost consequence: the wall’s per-tier Phase 1 obligation (one witness from the tier) is strictly cheaper than requiring intra-tier consensus. Mars’s three nodes are independent witnesses—any one satisfies the wall’s requirement. A Mars-internal majority would require two of three, adding a quorum obligation the wall does not need. The anchor tier is the sole exception: Phase 2 coherence requires it to behave as a single logical entity, achievable through internal replication (a locally replicated state machine). Non-anchor tiers benefit from redundancy without paying the cost of within-

tier agreement. The TLA+ verification provides independent evidence for this separation: the reduced 6-node topology ($3E + 1L + 1U + 1M$) is not merely a computational short-cut but a model of the collapsed-tier design—shrinking Earth from five nodes to three preserves the quorum intersection properties because the Phase 2 threshold rescales with it, holding the pigeonhole margin $|E| - k = 1$ fixed (under strict $\mathcal{Q}_2$, tier membership alone suffices). The same structural invariant that makes the reduction valid makes the deployment collapse cheap.

Topology-scoped consistency. During blackout, the wall separates tiers by which *capabilities* they retain. Earth, LEO, and Moon can still form both global quorums: they can complete Phase 1 (which surfaces every committed value: any Phase 1 quorum intersects every Phase 2 quorum) and Phase 2 (extend the global log), and, because they reach all five Earth nodes, they can assemble the acceptance evidence from a complete Phase 2 quorum that certifies a value as chosen. Under the standard Paxos client discipline, their operations on the global log remain linearizable. Mars retains neither capability: there are Earth-committed values Mars cannot discover until connectivity resumes, so it is confined to reading its local replica—a stale local view of the global history, not guaranteed to be a contiguous prefix—and to its separate Mars-local consensus scope. The capabilities are distinct, and the sparse topology of Section 6.3 separates them sharply: sparse LEO can still complete Phase 1, which surfaces accepted values, but *knowing* that a value is chosen requires acceptance evidence from a complete Phase 2 quorum—all five Earth nodes under the strict family—and sparse LEO reaches only three. Phase 1 alone buys visibility, not knowledge: LEO can neither extend the global log nor certify its contents, and is left observing values whose commitment it cannot confirm. The client-facing consequence is concrete: a tier holding both capabilities serves linearizable operations through the global log; a tier that can still assemble certification evidence but not complete an election can serve committed—if stale—reads; an observe-only tier like sparse LEO can return values only under an explicitly uncertified contract, since what it observes may yet be superseded; and a fully cut-off tier is confined to its local scope. The wall determines capabilities; precisely which consis-

tency contract to expose over a degraded capability set—and how the local scope reconciles on reconnection—is a service-level decision the wall does not make. What it makes legible is the capability boundary itself, and hence when that decision is being made: refusing operations at a cut-off tier preserves consistency by sacrificing availability, while continuing to serve locally is an explicit adoption of a weaker, local contract. The degradation is epistemic before it is electoral: what a cut-off tier loses first is not a vote but the ability to *know* what has been committed—chosenness is certified only by a complete Phase 2 quorum’s acceptances, and the wall together with the connectivity state bounds which tiers can still assemble that certificate. At interplanetary distances, the gap between the global and local regimes is twenty-two minutes wide—the distinction that distributed systems students struggle with as an abstract definition becomes a physical fact you could set a timer by.

Leadership hierarchy. The wall imposes a cost gradient on Multi-Paxos leader election that symmetric quorum constructions cannot express. Figure 3 shows the number of valid Phase 1 quorums per tier. The counts admit a closed form: the tier- i family contains $N_i = 2^{f_i} \prod_{j \leq i} (2^{|T_j|} - 1)$ quorums, where each required tier contributes a nonempty subset and the f_i nodes above tier i are unconstrained— $992 = 31 \cdot 2^5$ for Earth down to $217 = 7 \cdot 31$ for Mars. The exhaustive TLA+ enumeration confirms these counts rather than defining them. The closed form also makes the monotonicity a lemma rather than an observation: raising the initiator by one tier changes exactly one factor, tier T_{i+1} ’s contribution, from unconstrained ($2^{|T_{i+1}|}$) to nonempty ($2^{|T_{i+1}|} - 1$), so $N_{i+1} = N_i \cdot (2^{|T_{i+1}|} - 1) / 2^{|T_{i+1}|} < N_i$ for every wall, independent of tier sizes; the sequence $992 \rightarrow 496 \rightarrow 248 \rightarrow 217$ instantiates the ratios $\frac{1}{2}, \frac{1}{2}, \frac{7}{8}$. The argument survives relaxed Phase 2 unchanged, since the Earth-minimum factor is identical for every initiator tier and cancels from the ratio. The families are moreover nested—every tier- $(i+1)$ quorum is also a tier- i quorum—so any crash pattern under which a higher tier can still elect is one under which every lower tier can too; the counts quantify the gap in valid quorum sets, not availability under a stochastic failure model. In Multi-Paxos, where leader election *is* Phase 1, this makes Earth leadership both faster (sub-second vs. minutes) and strictly more crash-pattern-tolerant than Mars. The gradient is expressible only because the families are per-tier: any construction with a single initiator-independent Phase 1 family assigns every tier the same count by definition, which is the precise sense in which symmetric constructions cannot express it.

WPaxos [1] uses symmetric grid quorums with uniform Q_1 structure regardless of initiator. By contrast, the wall uses tier-indexed families $\mathcal{Q}_1^{(i)}$, so partition failover cost is topology-proportional: if the leader is on Earth and Mars disconnects, leader election remains Earth-scoped.

Terrestrial mapping. The separation principle applies

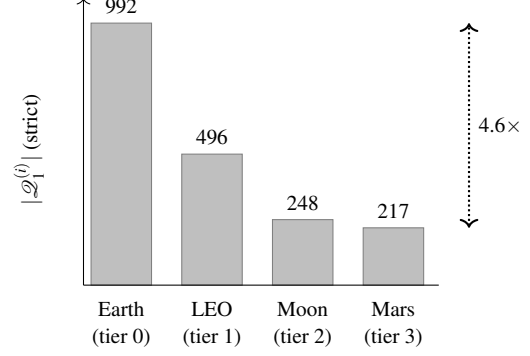


Figure 3: Phase 1 quorum count by initiating tier (strict $\mathcal{Q}_2$, from exhaustive TLA+ enumeration). Leadership flexibility increases monotonically from top to bottom of the wall.

wherever topology creates structured latency asymmetry. Consider a 3-tier edge deployment: T_0 = cloud region (one logical anchor, internally replicated across availability zones), T_1 = metro edge (2 retail-store servers), T_2 = remote site (1 offshore-platform node). Phase 2 anchors at the cloud; each tier’s Phase 1 reads downward. The cloud’s internal replication is its own concern; the edge servers are independent witnesses, any one of which satisfies the wall’s per-tier obligation. During planned WAN maintenance that disconnects the remote site, the wall predicts the capability matrix directly: cloud and metro edge retain both capabilities (they complete both phases and reach the anchor’s acceptance evidence) and continue serving linearizable operations; the remote site retains neither and is confined to its local state—the same states, read the same way, as Earth and Mars during conjunction. A metro-edge server whose maintenance window severs part of its cloud connectivity would enter sparse LEO’s observe-only state, and the wall says so before the window opens. Existing edge/cloud systems already separate these concerns in practice—cloud databases replicate within a region, edge nodes sync via ad hoc protocols—but they do so outside the quorum design, with bespoke consistency layers rather than structural guarantees. The wall gives that informal practice a formal foundation. Nor is the class limited to edge computing: any network with scheduled or predictable disconnection has the same shape—disconnected, intermittent, low-bandwidth (DIL) military networks, disaster-response deployments operating through WAN loss, and submarine or polar installations reachable only through windowed links. The interplanetary setting makes the separation undeniable; these deployments would benefit from the same decomposition made explicit. We state the status of this mapping plainly: it is an argument from structure, not an evaluated result—every claim in this paragraph is a hypothesis the wall makes precise, and validating them on terrestrial topologies is future work.

Future work. The separation principle suggests three nat-

ural extensions: scoped write leases, where the anchor tier delegates write authority to upper tiers over bounded slot ranges—lease expiry on disconnection, conflict resolution via MRDTs/CRDTs—so that topology-scoped consistency becomes a managed contract rather than a binary cutoff; structured multi-anchor families that trade part of the linear readout for tolerance of anchor-tier partition (Section 9); and empirical validation of the terrestrial mapping on edge/cloud topologies where the same decomposition applies but the latency asymmetry is less dramatic.

10 Conclusion

Flat quorums conflate inter-tier obligation with intra-tier replication. The wall-derived construction separates them: one witness per tier on the downward path as participation policy, with safety carried by the anchor row, while each non-anchor tier’s internal replication stays outside wall geometry. This yields legible liveness, a leadership cost gradient that symmetric constructions cannot express, and composable fault tolerance. Legibility extends to semantics: the wall and connectivity state reveal whether each tier can still learn and extend the globally ordered log or is confined to its local replica—the capabilities that determine the available consistency contract become a readable property of structure and connectivity rather than a hidden runtime condition.

Over a 10-node interplanetary topology, we confirm that three of four tiers retain global consensus during hard Mars blackout, at latencies determined by the speed of light to Earth. The extreme asymmetry makes the conflation error in flat quorum design undeniable, but the separation principle applies wherever topology creates structured asymmetry—the wall gives a formal foundation to practices that edge/cloud systems already follow informally.

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A Claim-to-Artifact Traceability

Table 7 maps the paper’s main claims to executable artifacts. Paths are relative to the repository root of the (anonymized) artifact accompanying this submission. Results were generated with Python ≥ 3.14 and SimPy $\geq 4.1.1$.

Table 7: *Claim-to-artifact traceability.*

Claim	Evidence	Artifact(s)
Flat construction: 0% during blackout at every hard-blackout sweep point.	Flat-mode sweep under the validated harness (-global-quorum flat).	results/step9_flat/step9_sweep.csv; experiments/step9_sweep.py.
Majority baseline: 100% during blackout, 361 ms two-phase decision latency at every hard-blackout sweep point.	Majority-mode sweep under the validated harness (-global-quorum majority).	results/step9_maj/step9_sweep.csv; experiments/step9_sweep.py.
Crumbling wall: 100% during blackout at every hard-blackout sweep point, 183 ms two-phase decision latency.	All rows show 100% during-blackout.	results/step9/step9_sweep.csv; experiments/step9_sweep.py.
Per-tier liveness: 3 of 4 tiers retain global consensus during blackout.	Per-tier sweep, both topologies.	results/tier_liveness/ tier_sweep_full_ci.csv; experiments/ tier_liveness_sweep.py.
Sparse topology: LEO drops to 0%.	Sparse rows in per-tier sweep.	Same as above.
Weakest-link migration and coordinated relaxation.	Crash-tolerance sweep rows.	results/step10/step10_sweep_ci.csv; experiments/step10_sweep.py.
TLA+ verification of quorum intersection.	Exhaustive enumeration, all four tiers $\times$ {strict, $k=4$, $k=3$ } (27,921 states, complete).	tla/QuorumIntersection.tla; tla/ ExhaustiveIntersection.tla.
Paxos agreement model-checked.	67M states (reduced topology, all-tier Phase 1 family, complete).	tla/PaxosSmall.tla.
Per-tier capability map: which (Phase 1, Phase 2) capability states each tier holds under each connectivity regime.	Exhaustive per-scenario classification, including the degraded partial-capability states.	results/capability/capability_map.csv; experiments/capability_map.py.